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基於大型語言模型多代理人的社會影響動態模擬

Simulating Social Influence Dynamics with LLM-based Multi-agents

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基於大型語言模型多代理人的社會影響動態模擬
Simulating Social Influence Dynamics with LLM-based Multi-agents

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摘要

模擬人類社會行為一直是計算社會科學中的核心挑戰。傳統的代理人基模型雖可提供受控環境以探索群體現象，然而由於代理人行為邏輯簡化、互動規則僵化，其模擬之行為真實性受限。本研究旨在探討以大型語言模型驅動之多智能體系統，是否能更真實地再現諸如從眾效應、群體極化與意見分裂等社會影響動態。研究設計採用具備語言流暢度與社會規範理解能力的大型語言模型，建構基於非同步文字交流的模擬架構，仿照電子佈告欄平台之互動形式。本研究系統性操控模型規模、架構特徵與推理模組能力，並提出量化指標以衡量立場轉變、群體極化程度與共識形成趨勢。實驗結果顯示，大型語言模型不僅能生成具社會行為特徵的互動，且其社會影響敏感度與立場穩定性會隨模型架構不同而顯著變化。本研究在方法上為模擬人類群體行為提供新途徑，亦對未來社會感知型多代理系統之設計具參考價值。

關鍵詞：大型語言模型、社會模擬、多代理人系統、基於大型語言模型之多代理人、社會影響

Abstract

Simulating human social behavior has long been a central challenge in computational social science. While classical agent-based models offer controlled environments for exploring group-level phenomena, their behavioral realism remains limited due to simplified agent logic and deterministic interaction rules. This study investigates the potential of LLM-based multi-agent systems to more accurately emulate complex social influence dynamics, including conformity, polarization, and fragmentation. By leveraging LLMs' linguistic fluency and implicit knowledge of social norms, we construct a structured simulation framework grounded in asynchronous, text-based bulletin board interactions. Our empirical design systematically varies model scale, architectural features, and reasoning capabilities to assess their impact on emergent group behavior. We introduce quantitative metrics to capture stance shifts, group divergence, and alignment trends across multi-round discussions. Results reveal that LLMs not only replicate recognizable social behaviors but also exhibit divergent influence sensitivity depending on their cognitive architecture. This work contributes methodological insights for simulating human-like group dynamics and offers practical guidance for the design of socially aware multi-agent systems.

Keywords: Social Simulation, Large Language Model, Large Language Model-Based Multi-agents, Multi-Agent Systems, Social influence

Table of Contents

論文審定書 ..... **i**

摘要 ..... **ii**

Abstract ..... **iii**

Table of Contents ..... **iv**

List of Figures ..... **vi**

1. Introduction **1**

2. Background **5**

2.1 Social Influence Phenomena in Group Dynamics 5

2.2 Agent-Based Modeling: Strengths and Limitations 6

2.3 Large Language Models as Enhanced Agent Architectures 7

2.4 Forum-Based Environments for Social Interaction Modeling 8

2.5 Selecting a Multi-Agent Orchestration Framework 9

2.6 Empirical Advances in LLM-Based Multi-Agent Simulations 10

2.7 Methodological Positioning and Research Contributions 11

3. Methodology **13**

3.1 Agent Architecture and Persona Design 13

3.2 Model Classification and Design 14

3.3 Experimental Design and Replication 15

3.4 Evaluation Metrics 17

3.4.1 Conformity Rate Measurement 18

3.4.2 Group Polarization Index 19

3.4.3 Group Fragmentation Index 20

4. Experiment and Discussion 22

4.1 Single-Model Discussion Results..... 22

4.2 Intra-Group Discussion Results.....25

4.3 Cross-Group Discussion Results..... 27

4.4 Implications for Multi-Agent System Design 29

5. Conclusion 31

Reference 33

List of Figures

Figure 1. Architecture for the Conversation Simulation

Figure 2. Conformity Rates in Single-Group Discussions.

Figure 3. Agent Stance Trends in Single-Group Discussions.

Figure 4. Conformity Rates in Intra-Group Discussions.

Figure 5. Agent Stance Trends in Intra-Group Discussions.

Figure 6. Conformity Rates in Cross-Group Discussions.

Figure 7. Agent Stance Trends in Cross-Group Discussions.

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1. Introduction

Classic human subject experiments have established foundational principles of social behavior that continue to shape research across psychology, governance, healthcare, and economics. The Asch Conformity Experiment revealed that individuals often adjust their accurate perceptions to align with group consensus, highlighting the role of peer pressure in shaping decision-making under social uncertainty (Asch, 1955). Milgram's Obedience Study demonstrated how individuals may comply with authoritative commands even when those actions conflict with personal moral judgment, underscoring the power of hierarchical influence (Milgram, 1963). These and other seminal studies have provided enduring insights into the dynamics of collective behavior across institutions, policymaking processes, and online communities (Cau et al., 2025). At their core, these experiments illustrate that human social responses, though context-dependent, follow structured patterns that can be systematically investigated through controlled empirical methods.

Traditional human subject experiments remain indispensable for understanding social behavior, yet they face persistent limitations in cost, scalability, and ethical feasibility (Squazzoni et al., 2014). Recruiting participants, managing long-term studies, and manipulating socially sensitive variables under controlled conditions can be both resource-intensive and ethically constrained. These challenges have motivated the exploration of alternative methods that preserve behavioral realism while allowing for systematic experimentation.

LLM-based multi-agent simulations (Guo et al., 2024) present a promising complement to this landscape. Trained on extensive corpora of human-generated discourse, these models internalize not only linguistic conventions but also behavioral

tendencies and social norms, enabling them to generate plausible, context-aware responses to conversational stimuli. When deployed in multi-agent settings, LLMs can engage in dynamic exchanges that reflect recognizable patterns of group interaction, including peer influence, alignment with majority views, and persistence of dissenting perspectives.

This convergence of computational modeling and empirical social science raises a central research question: To what extent can simulations built from LLMs serve as scalable, ethically tractable proxies for studying group-level behavioral dynamics traditionally explored through human experiments (Piao, Yan, et al., 2025)? Addressing this question allows us to assess the validity of LLM-based agents as tools for modeling how individuals negotiate belief, identity, and stance within social contexts, potentially expanding the methodological repertoire of computational social science.

The study of LLM-driven social interaction necessitates an experimental environment that emphasizes text-based exchanges, accommodates asynchronous participation, and ensures transparent tracking of how positions evolve, while limiting potential confounds such as visual cues or real-time conversational pressure. Bulletin Board System (BBS) platforms offer a suitable match for these requirements by enabling structured, message-based interaction across multiple contributors in staggered turns. Within this setting, participants engage in extended discussions where viewpoints are gradually shaped through repeated exposure and response cycles, often giving rise to opinion adjustments, emotional reinforcement, and collective alignment. The format also promotes deliberation: individuals articulate arguments, contest opposing claims, and refine their stances across successive rounds of dialogue. As a result, BBS-style environments provide a controlled yet ecologically valid context for examining whether

LLM-based agents are capable of replicating the dynamics of social influence that emerge in human groups.

This research contributes to both the field of agentic artificial intelligence and computational social science by introducing a set of methodological refinements. We develop and assess a structured multi-agent dialogue framework designed to emulate the asynchronous communication style typical of BBS platforms. Building on this foundation, our empirical analysis is organized around three central research questions:

1. **Social Behavior Emulation:** How do LLM-based multi-agent simulations exhibit social influence dynamics such as conformity, polarization, and fragmentation?
2. **Model Capacity Effects:** How do differences in LLM parameter scale and architectural design influence the emergence and intensity of group-level dynamics, including stance volatility and convergence?
3. **Reasoning Module Impact:** What role do specialized reasoning capabilities play in shaping agents' resistance to peer influence, stance retention, and contribution to group-level opinion structure?

This work advances the domains of agentic artificial intelligence and computational social science by introducing methodological contributions that enable more realistic simulations of group interaction. We design and evaluate a multi-agent conversational framework that mirrors the asynchronous communication flow characteristic of BBS-style platforms. Within this framework, our experiments systematically vary model scale and reasoning capacity to assess their effects on interactional dynamics, measuring indicators such as conformity, polarization, and stance persistence across multiple rounds of discussion. The findings demonstrate clear associations between model architecture and emergent social influence patterns, validating the framework's capacity to capture human-like conversational behaviors.

Beyond empirical insights, the study offers methodological guidance for future investigations and practical implications for developing socially aware multi-agent systems.

The structure of this thesis is organized as follows. Section 2 surveys the existing literature, outlining foundational concepts in social simulation, the role of large language models, and previous attempts to capture human-like behavior through LLM-based agents. Section 3 introduces the methodological framework, describing the system architecture, agent persona design, and the metrics employed for evaluation. Section 4 reports the experimental setup and findings, combining quantitative measures with qualitative observations to analyze emergent social dynamics. Section 5 concludes by highlighting the principal results and discussing their broader implications.

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2. Background

The theoretical foundation for LLM-based social simulation requires the examination of established social psychological principles, computational modeling approaches, and the technical infrastructure necessary for implementing multi-agent interactions. This section systematically reviews these components to establish the conceptual framework underlying our experimental design.

2.1 Social Influence Phenomena in Group Dynamics

Group behavior and decision-making processes represent fundamental research areas spanning social psychology (Zhang et al., 2024), communication theory, and computational social science. Among the foundational phenomena in this domain is conformity, which refers to the tendency of individuals to adjust their beliefs or actions to align with the prevailing norms or opinions of a group. This behavioral adaptation is particularly pronounced in contexts of social uncertainty or peer pressure, where the cost of deviation may outweigh the perceived benefit of holding an independent stance. Classic experiments such as Asch's line judgment study have demonstrated how individuals override their private judgments to conform to group consensus, even when that consensus is objectively incorrect (Weng et al., 2025).

Another well-documented phenomenon is group polarization, wherein individuals with initially moderate or mixed viewpoints shift toward more extreme positions following repeated group interactions. This dynamic is often fueled by exposure to like-minded opinions, confirmation bias, and normative pressure within homogenous subgroups (Cau et al., 2025; Piao, Lu, et al., 2025). As a result, discussions tend to amplify ideological positions rather than moderate them, contributing to increased divergence between opposing factions. Polarization has been observed in both offline deliberative

settings and online communities, where algorithmic content filtering and social signaling exacerbate the reinforcement of extreme views (Chuang et al., 2024).

Fragmentation, in contrast, describes the structural separation of a group into persistent, ideologically opposed subgroups that lack communicative bridges or consensus pathways. Unlike polarization, which can occur within a single spectrum of opinion, fragmentation involves the formation of distinct, non-overlapping clusters of belief. This phenomenon reduces the likelihood of dialogue, increases mutual distrust, and can undermine collective decision-making processes (Flache et al., 2017; Qasmi et al., 2025). Taken together, conformity, polarization, and fragmentation form the conceptual core for analyzing how social influence shapes group-level dynamics in both human and simulated systems.

Collectively, these three phenomena represent the most widely recognized dimensions of social influence. Conformity captures individual-level susceptibility to majority pressure, with extensive experimental evidence showing its impact on decision-making. Polarization reflects group-level dynamics where discussions amplify existing tendencies and push opinions toward the extremes, a pattern consistently observed in both offline deliberations and online platforms (Jost et al., 2022; Liu et al., 2023). Fragmentation highlights structural outcomes at the societal level, where populations split into persistent, opposing subgroups, undermining cohesion and consensus (Minh Pham et al., 2020; *The Group Effect*, n.d.). By spanning the individual, group, and structural levels, these phenomena provide a comprehensive analytical lens for understanding social influence. For this reason, they have been chosen as the central metrics in our simulation, ensuring that the evaluation is grounded in well-established theoretical and empirical research rather than in arbitrary design choices.

2.2 Agent-Based Modeling: Strengths and Limitations

Traditional approaches to studying group behavior and social influence have primarily relied on a combination of controlled laboratory experiments, field observations, and survey methodologies. These human-centric methods have enabled researchers to systematically investigate the mechanisms underlying group decision-making, social pressure, and behavioral adaptation (Aher et al., 2023). However, they often suffer from challenges related to cost, scalability, and experimental control, particularly when examining complex, multi-agent interactions in naturalistic settings. To address these constraints, computational modeling techniques emerged as complementary tools, offering a platform to simulate group behaviors in silico. Among these, agent-based modeling (ABM) has played a foundational role by allowing researchers to define agent-level decision rules and observe emergent group-level phenomena across diverse social scenarios (Squazzoni et al., 2014).

Despite its methodological utility, classical ABM is limited in its capacity to authentically replicate human behavioral complexity. Traditional ABM agents operate based on predefined heuristics and deterministic rules, which, while useful for exploring theoretical dynamics, fail to capture the adaptive reasoning, linguistic subtlety, and contextual awareness inherent to real-world social interactions (L. Wang et al., 2025). As a result, ABM-generated behaviors often appear mechanistic or stylized, lacking the communicative richness and psychological nuance that characterize human group behavior. These shortcomings have motivated the exploration of more expressive agent architectures, particularly those powered by Large Language Models (LLMs), which can simulate intricate conversational exchanges and dynamic belief updates. While ABM remains valuable for abstract system-level modeling, its limitations underscore the need

for enhanced simulation frameworks capable of bridging the gap between theoretical formalism and human-like interaction patterns (Leng & Yuan, 2024).

2.3 Large Language Models as Enhanced Agent Architectures

Large Language Models (LLMs), have significantly expanded the design space of agent-based modeling by endowing agents with powerful natural language processing capabilities and access to vast textual knowledge. These models are trained on diverse internet-scale corpora, including online forums, social media conversations, and news articles, which enable them to acquire syntactic fluency, pragmatic nuance, and embedded social conventions (Jiang et al., 2024).

Beyond linguistic fluency, LLMs demonstrate capabilities that closely approximate aspects of human social cognition. When deployed as autonomous conversational agents, they can dynamically interpret social cues, express opinions with coherence, and revise stances in response to peer influence. These behaviors emerge from the models' ability to simulate mental-state reasoning, track dialog history, and apply implicit norms derived from training data. For example, LLM-based agents can exhibit conformity to majority viewpoints, reinforce extreme attitudes through repeated exposure, or adaptively align with evolving group consensus—all phenomena characteristic of human group interactions (Jiang et al., 2024). These features position LLMs as promising agent architectures for simulating social dynamics with a level of behavioral richness and flexibility that traditional ABM agents lack.

2.4 Forum-Based Environments for Social Interaction Modeling

Studying LLM-based social interaction requires an experimental environment that supports asynchronous, text-based exchanges while minimizing extraneous variables such as visual bias or real-time pressure. Bulletin Board System (BBS) platforms offer a

suitable setting for this purpose. These systems facilitate structured, temporally staggered communication, enabling participants to engage in multi-round discussions without the cognitive demands of synchronous interaction. The BBS format allows for detailed tracking of message flow, stance progression, and interaction dependencies, making it an ideal substrate for controlled simulations of opinion dynamics. By removing confounding factors present in spoken or real-time modalities, BBS environments isolate textual influence as the primary vector of social alignment and disagreement (Aher et al., 2023; Qasmi et al., 2025).

Beyond their technical affordances, BBS platforms foster social behaviors essential for studying group-level phenomena. Participants in such environments routinely express opinions, respond to opposing viewpoints, and occasionally revise their positions in light of repeated exposure or persuasive argumentation. This iterative structure supports emergent phenomena such as stance shifts, emotional reinforcement, and alignment with perceived group norms. Moreover, the reflective pacing of BBS interactions allows agents—human or artificial—to process previous content and generate context-sensitive responses, closely mirroring the deliberative nature of real-world online forums. Consequently, BBS forums provide a realistic yet experimentally tractable setting to evaluate whether LLM-based agents can authentically replicate social influence mechanisms observed in human discussions.

2.5 Selecting a Multi-Agent Orchestration Framework

Implementing multi-agent simulations using LLMs requires a dedicated orchestration framework that supports structured communication, flexible agent role assignment, and reproducible interaction flows. Several frameworks have recently emerged to meet these demands, including LangGraph (*LangGraph*, n.d.), CrewAI (*Introduction*, n.d.), and AutoGen (Wu et al., 2023). These toolkits offer varying degrees

of abstraction and control, from declarative state-machine workflows to fully customizable conversation engines. LangGraph, for instance, enables graph-based planning and memory-persistent agent interactions, while CrewAI emphasizes role specialization and task chaining. Given the experimental need for turn-based, forum-style conversations with precise agent coordination, selecting the appropriate orchestration platform is a non-trivial design decision.

Among the available options, AutoGen was selected for this study due to its robust support for multi-round dialogue management, persona maintenance, and flexible message routing. Its modular architecture enables researchers to simulate both synchronous and asynchronous discussion settings while preserving interaction history across turns. Crucially, AutoGen provides fine-grained control over conversation flow, allowing agents to reference specific messages, respond to targeted individuals, and update positions over time. These capabilities make it particularly well-suited for emulating BBS-style discussions in controlled experiments, where the traceability of stance evolution and inter-agent referencing is essential. Furthermore, its logging and session management tools facilitate systematic data collection for post-hoc analysis of group behavior dynamics.

2.6 Empirical Advances in LLM-Based Multi-Agent Simulations

A growing number of studies have explored the use of LLMs in simulating social phenomena within multi-agent settings. For example, Gao et al. demonstrated that GPT-based agent networks can emulate cooperative behaviors and decision-making coordination in digital communities, revealing dynamics reminiscent of human collective learning (Gao et al., 2023), (Helbing, 2012). Park et al. extended this line of work by constructing persistent agent personas capable of maintaining memory and evolving through interaction in virtual environments, offering insights into group-level behavioral

emergence (Li et al., 2023). Additional studies have investigated negotiation, coalition formation, and role differentiation among LLM agents, providing early evidence that language-based agents can generate structured, socially coherent behaviors across multiple conversational rounds (Chuang et al., 2024).

While these studies represent significant steps forward, they also expose notable limitations that motivate the present research. Many existing simulations lack empirical validation against human baselines, raising questions about the realism of LLM-generated group behavior (Cau et al., 2025; Jiang et al., 2024). Moreover, most investigations rely on a single model or architecture, limiting our understanding of how model capacity, reasoning ability, or training variation shape social dynamics during extended interactions (Mou et al., 2024). Finally, there remains no standardized approach to quantifying social influence phenomena such as conformity, polarization, or fragmentation in these systems, making it difficult to compare findings across studies or establish theoretical grounding for observed behaviors. These shortcomings highlight the need for a more systematic, comparative, and metrics-driven investigation—one that this study seeks to address.

2.7 Methodological Positioning and Research Contributions

This study seeks to close important gaps in prior research by examining, under controlled experimental settings, how differences in LLM scale and architectural design shape the dynamics of BBS-style simulated discussions. Drawing on principles from social psychology and integrating them with recent progress in agent-oriented AI, the work advances understanding of the extent to which artificial agents can reproduce complex patterns of group behavior. Unlike earlier studies that largely emphasized proof-of-concept demonstrations or analyses limited to a single model, our approach highlights cross-model comparison and behavioral fidelity, clarifying the conditions under which simulated dialogues reflect established social influence phenomena.

Extending from these considerations, we present a structured experimental framework to evaluate conformity, polarization, and fragmentation in LLM-based multi-agent systems. The framework incorporates standardized conversational protocols, stance-monitoring procedures, and quantitative measures that allow systematic comparison across different model designs. Such a methodological setup enables rigorous assessment of simulation reliability, ensures reproducibility, and reveals the architectural characteristics most influential to behavioral realism. Collectively, these contributions provide practical direction for future research in computational social science and inform the development of socially adaptive multi-agent systems for both scholarly inquiry and real-world applications.

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3. Methodology

We develop a conversational environment based on LLM-driven multi-agent systems to emulate human-style exchanges within BBS-like platforms. The architecture is coordinated by a central controller that manages turn-taking through a round-robin sequence, where each agent contributes messages in a predefined order and all outputs are shared across the group. This arrangement allows agents to engage with earlier contributions, quoting or responding to prior messages like interactions observed in real-world forum threads.

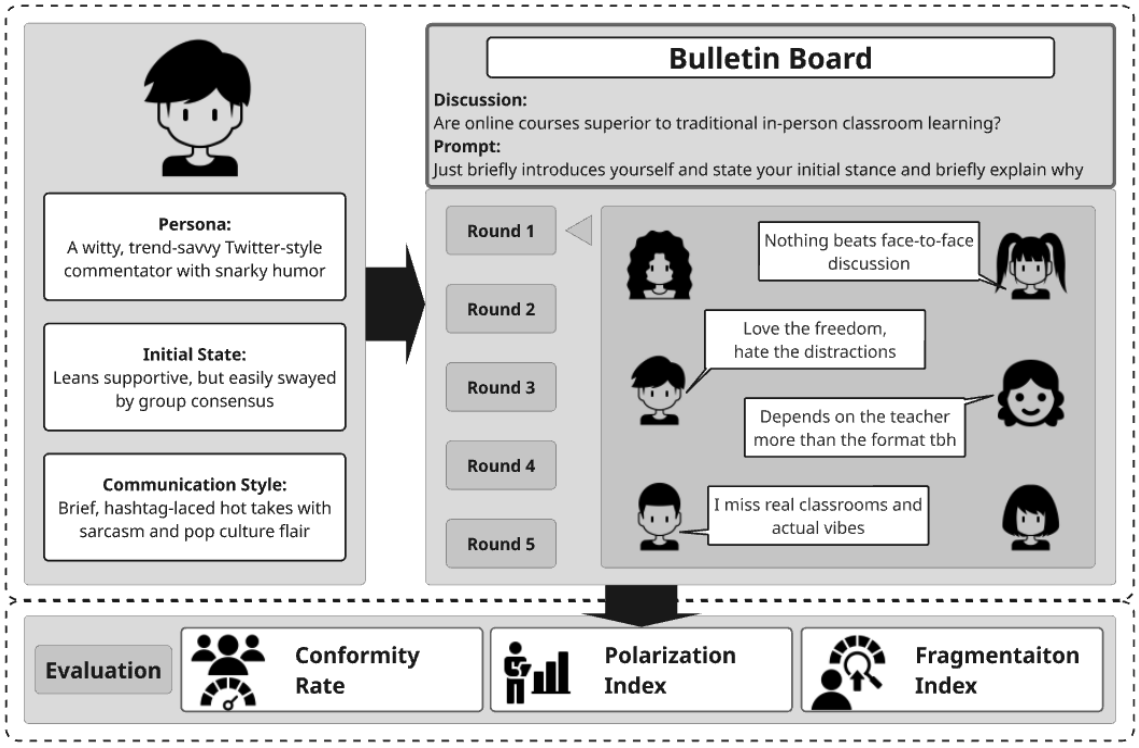


Fig. 1. Architecture for the Conversation Simulation

3.1 Agent Architecture and Persona Design

Each agent is initialized with a structured persona prompt that specifies demographic traits, communication style, and an initial stance on the discussion topic. This configuration maintains consistent starting conditions across different LLMs (Z.

Wang et al., 2024), allowing the effects of model architecture on social behavior to be examined in isolation. For example, one role may represent an idealistic and open-minded participant who supports a proposal unless strong counterarguments are presented. The complete schema for persona construction, together with the procedural workflow of multi-agent exchanges, is illustrated in Fig. 1, which depicts the bulletin-board style simulation environment.

The interaction protocol spans five sequential rounds of text-based contributions. In the opening round, a central coordinator introduces a contentious issue (e.g., whether governments should adopt strict environmental regulations). Each agent then provides an initial statement aligned with its persona characteristics and assigned stance. In subsequent rounds, agents are instructed to reference or reply to earlier contributions, fostering engagement while preserving their designated communicative style. This design reproduces asynchronous turn-taking, ensuring equal participation opportunities and allowing agents to adjust or maintain their views as the discussion progresses. By the end of Round 5, every agent will have generated five contributions, producing a complete dialogue log that can be analyzed for stance evolution and rhetorical dynamics.

3.2 Model Classification and Design

The study employed four groups of large language models, each containing three distinct models. The grouping strategy was guided by architectural family, parameter scale, and reasoning capability, with the aim of capturing systematic contrasts in how different classes of models shape collective dynamics. No single model was sufficient to satisfy all desiderata of openness, performance, and accessibility. Consequently, the design relied on selecting clusters of models that were either from the same series or exhibited comparable functional capacities.

1. Group A includes models that operate efficiently on a single GPU, offering a balance between accessibility and core language understanding capabilities (e.g., Qwen2.5-7B (Qwen et al., 2025), Llama3.1-7B (Grattafiori et al., 2024), and Deepseek-R1-8B (DeepSeek-AI et al., 2025)).
2. Group B encompasses larger models that provide increased capacity while remaining deployable in constrained computational environments (e.g., Qwen2.5-72B (Qwen et al., 2025), Llama3.1-70B (Grattafiori et al., 2024), and Deepseek-R1-70B (DeepSeek-AI et al., 2025)).
3. Group C consists of proprietary LLMs with broad deployment and optimized performance, such as GPT-4o (OpenAI et al., 2024), Claude 3.5 Haiku (yet, n.d.), and Gemini Flash 2.0 (G. Team et al., 2024).
4. Group D comprises models specifically designed or fine-tuned for enhanced logical reasoning and inference, including GPT-o1-mini (Zhong et al., 2024), Deepseek-R1 (reasoning variant) (DeepSeek-AI et al., 2025), and QwQ-32B (Q. Team, 2024).

All groups were evaluated under a unified experimental protocol, with consistent persona assignments and prompt structures. This design minimizes external confounds and enables clear attribution of behavioral differences to variations in model scale, architecture, and reasoning orientation. The setup supports a systematic exploration of how these factors influence agent-level stance shifts and emergent group dynamics.

3.3 Experimental Design and Replication

To mitigate the variability inherent in LLM outputs and strengthen the reliability of findings, each simulation was executed as a full five-round, turn-taking discussion and replicated 25 times independently. Every run included the entire sequence from Round 1 through Round 5. Results from these repetitions were then aggregated to capture

consistent patterns in conformity, polarization, and fragmentation. This replication protocol ensured that emergent behaviors reflected stable tendencies rather than single-run anomalies.

Each dialogue was limited to five rounds. This decision was based on empirical observation that, within this interval, agent stances frequently displayed convergence or stabilization. While no prior literature prescribes an exact number of rounds for LLM-based deliberations, five iterations offered a balance between tractability and interaction depth. Fewer rounds would truncate potential influence cascades, while additional rounds yielded diminishing changes in stance trajectories. This pragmatic choice allowed the simulation to capture meaningful influence dynamics without unnecessary computational overhead.

The discussions followed a structured, turn-taking format rather than an asynchronous free-flowing exchange. In each round, agents contributed sequentially, ensuring equal participation opportunities and a transparent trajectory of stance evolution. This format was selected to maximize comparability across runs, as it eliminates confounding factors such as message overlap or uneven participation frequency. In contrast, real-world forums allow asynchronous posting, which may accelerate echo chambers or amplify polarization by enabling selective reinforcement. While the structured format reduces spontaneity, it provides a controlled environment for systematically analyzing conformity, polarization, and fragmentation.

1. **Single-Model Discussion (Baseline):** In this configuration, all six agents within the simulation are instantiated from the same LLM. This provides a controlled baseline for observing group dynamics in homogeneous environments, where any differences in stance evolution or conformity originate solely from the stochastic variability of generative outputs.
2. **Intra-Group Multi-Model Discussion:** Six agents were instantiated from three distinct LLMs within the same predefined family. Because each family contained

only three models, each model was duplicated and randomly assigned to two different agent roles. To mitigate role bias, model-to-agent assignments were reshuffled across runs, ensuring that no model was confined to a fixed conversational position. This approach preserved intra-group diversity while enhancing robustness while preserving intra-group diversity.

3. **Cross-Group Multi-Model Discussion:** Agents were instantiated from across all four families, with at least one representative from each group included in every run. The two remaining slots were filled by randomly selected models drawn from the overall pool. The assignment was re-randomized for each trial, generating multiple distinct compositions. This maximally heterogeneous setting emulates pluralistic social contexts where agents embody divergent architectures, parameter scales, and reasoning strategies.

All configurations were independently executed 25 times. Randomization ensured that no individual model consistently occupied the same role, while repeated trials reduced the influence of outlier runs. Aggregated across replications, this procedure provided a reliable assessment of how conformity, polarization, and fragmentation emerged under homogeneous, intra-group, and cross-group conditions.

Collectively, this experimental design enabled a comparative evaluation of group behaviors across different levels of diversity. From homogeneous alignment in single-model settings, through intra-group cascades, to heterogeneous negotiation in cross-group simulations, the design offered a systematic lens for assessing how architecture and composition shape influence dynamics in LLM-based multi-agent systems.

3.4 Evaluation Metrics

Our analysis centers on three key social phenomena: conformity, polarization, and fragmentation. These measures were deliberately selected because they capture distinct yet complementary layers of social influence. At the individual level, conformity assesses whether an agent changes its stance in response to majority pressure. At the group level, polarization reflects whether collective discussion pushes opinions toward more extreme

positions. At the structural level, fragmentation evaluates whether the population divides into opposing camps rather than converging toward a shared consensus. Taken together, this triplet of indicators spans the continuum from individual behavior to group dynamics and overall structural outcomes. This design offers a more comprehensive perspective than any single measure, providing a stronger basis for assessing whether LLM-based simulations can exhibit recognizable patterns of social influence.

3.4.1 Conformity Rate Measurement

Conformity, within the context of group decision-making, describes the tendency of individuals to adjust their expressed opinions so that they align more closely with the prevailing majority stance (Helbing, 2012). In our simulation framework, each agent provides an initial declaration in Round 1 and then encounters four subsequent opportunities to either retain its position or revise it in response to the evolving discussion. A revision is categorized as conforming when the change decreases the distance between the agent's stance and the dominant group position at that round.

Formally, let N denote the total number of stance-change opportunities, defined by six agents each having four update windows, i.e., $N = 24$. The Conformity Rate (CR) is defined as:

$$CR = \frac{\sum_{i=1}^{N} \mathbf{1}\{\text{change } i \text{ aligns with majority}\}}{N}$$

In this formulation $\mathbf{1}\{\cdot\}$, the indicator function takes the value of 1 when the ith stance update moves the agent closer to the prevailing majority in that round, and 0 otherwise. A higher CR indicates greater susceptibility to majority influence, with agents frequently shifting their stance toward the prevailing opinion. A lower CR reflects resistance to peer pressure, implying that agents maintain their initial positions even in the presence of

opposing majorities. Values near zero suggest independence, while values approaching one indicate strong conformity pressure.

3.4.2 Group Polarization Index

Group polarization describes the process by which participants with initially diverse or moderate viewpoints gradually adopt more extreme positions after collective discussion. In this study, each agent's stance is encoded on a five-point ordinal scale: Strongly Oppose (−2), Oppose (−1), Neutral (0), Support (+1), and Strongly Support (+2). At the end of every round, the most recent stance of each agent is recorded, producing a distribution of opinions across the group.

To quantify the extent of polarization, the stance distribution at round r is represented as a probability vector, where $p_r(s)$ denotes the fraction of agents holding stance at round r. The Polarization Index P_r s then defined as the expected absolute value of stance:

$$P_r \;=\; \sum_{s=-2}^{+2} \left| s \right| \;\times\; p_r(s).$$

Since $|s| \in \{0, 1, 2\}$ the index is bounded between 0 (all agents remain neutral) to 2 (all agents are positioned at either extreme). This measure captures the extent to which opinions are distributed toward the extremes. For example, consider a stance at round r where:

$\{p_r(-2) = 0.166,\ p_r(-1) = 0.166,\ p_r(0) = 0.336,\ p_r(+1) = 0.166,\ p_r(+2) = 0.166\}$

, then $P_r = (2 \times 0.166) + (1 \times 0.166) + (0 \times 0.336)$ $+(1 \times 0.166) + (2 \times 0.166) = 0.83$. This value corresponds to a moderate spread of opinions, with agent stances distributed across the ideological spectrum rather than clustered at the center or extremes.

To measure how polarization evolves, we define the Polarization Change $\Delta P = \left[P_5 - P_1\right]$. representing the difference between the final and initial polarization indices. This metric captures the net shift in opinion extremity across five discussion rounds. For instance, if $P_1 = 0.83$ and $P_5 = 1.53$, then $\Delta P = 0.7$ indicates a substantive increase in polarization. Such changes are especially meaningful when baseline fluctuations are limited under alternative model configurations, suggesting that specific agent architectures may intensify group divergence over time.

3.4.3 Group Fragmentation Index

Group fragmentation characterizes situations where a population separates into clearly defined clusters with opposing orientations, instead of moving toward a shared consensus. In our simulation, fragmentation is operationalized as the simultaneous presence of two dominant opinion clusters at the extremes of the stance spectrum—namely, agents who Support or Strongly Support the proposition, and those who Oppose or Strongly Oppose it. Agents holding Neutral stances are treated as non-aligned and are not considered part of either faction.

To quantify this divergence, we define the Fragmentation Index F_r at round r as:

$$F_r = 1 - \left| p_r^{\text{support}} - p_r^{\text{oppose}} \right|$$

Where p_r^{support} denotes the proportion of agents expressing either Support (+1) or Strongly Support (+2), and p_r^{oppose} denotes the proportion expressing Oppose (−1) or Strongly Oppose (−2) at round r. The index F_r ranges from 0 (all agents on one side, no fragmentation) to 1 (equal representation of opposing sides, maximum fragmentation).

As an illustration, consider a case where 20% of the agents fall into the Strongly Support category and another 20% into the Strongly Oppose category, while the remaining 60% remain Neutral. The balance between the two extremes yields $F_r = 1$, signifying a highly fragmented state. In contrast, if one side dominates, such as 80% of agents supporting and only 10% opposing, the imbalance produces a lower fragmentation index, indicating that opposing factions are comparatively weak.

Our multi-agent simulation environment combines structured turn-taking, persona-grounded agent prompts, and standardized stance encoding to enable consistent behavioral tracking across experiments. These design choices allow for precise measurement of CRs, polarization indices, and fragmentation levels over multiple conversational rounds. The following section presents comparative results across different LLM configurations, highlighting how architectural differences influence the emergence of these social phenomena.

4. Experiment and Discussion

The experiments in this study were designed and executed under standardized conditions, adhering to the methodological framework outlined in Section III. All language models were exposed to identical prompt structures and conversational environments to facilitate direct comparison. Multi-agent interactions were coordinated using AutoGen (Wu et al., 2023), which enabled uniform management of persona-driven messaging flows across distinct model groups. The models were categorized into four primary types based on parameter size, hardware demands, and the presence or absence of explicit reasoning mechanisms. This experimental setup enabled a systematic investigation into how variations in model architecture and capacity influence emergent social behaviors, including alignment tendencies and stance progression.

4.1 Single-Model Discussion Results

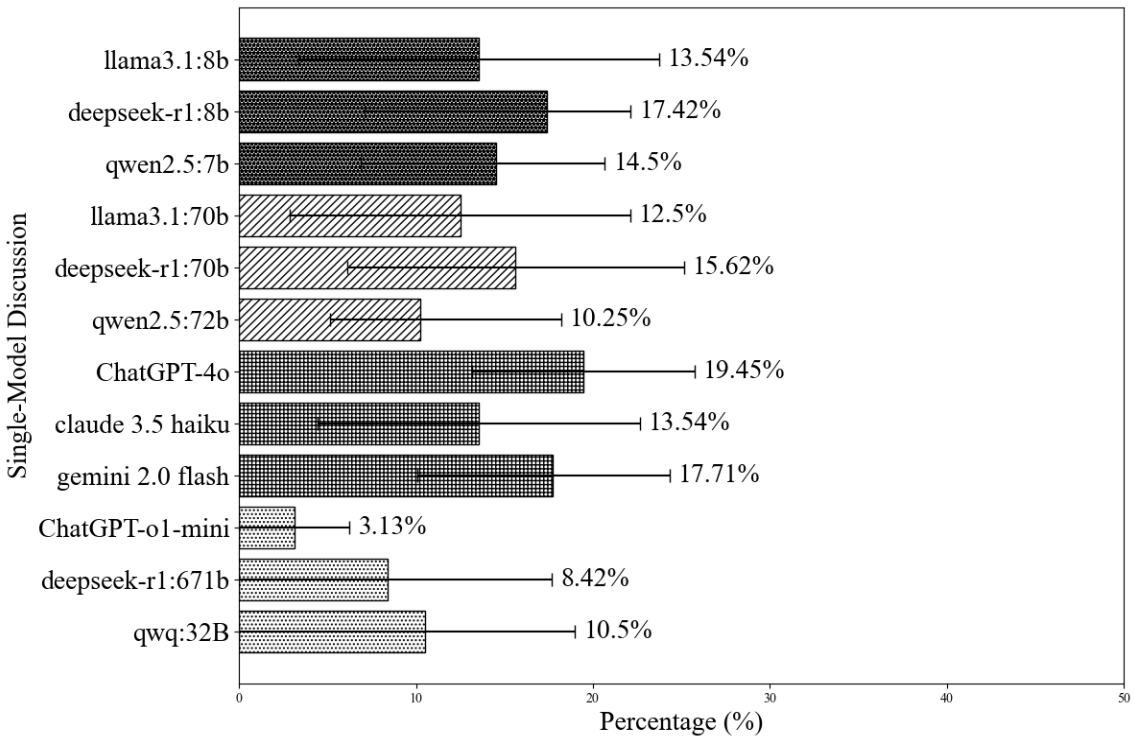


Fig. 2. Conformity Rates in Single-Group Discussions.

We begin with the homogeneous baseline to establish how uniform model composition shapes alignment pressure before introducing intra-group and cross-group heterogeneity. As shown in Fig. 2, the single-model baseline, six agents instantiated from the same LLM, exhibit a clear tendency toward majority alignment. Conformity concentrates at mid-to-high levels for most base generative models, consistent with a setting where agents share priors, rhetorical habits, and error modes. Notable outliers anchor the extremes: a reasoning-centric model sustains near-floor CR, while a proprietary generalist attains one of the highest CRs in this condition. Taken together, these patterns indicate that architecture matters more than scale for alignment pressure: parameter increases within the same family do not systematically reduce CR, whereas adding explicit reasoning scaffolds does. Because the protocol fixes personas, turns budgets, and rounds structure, the observed differences are best explained by inductive bias and decision-updating style rather than positional artifacts. Single-model homogeneity effectively amplifies normative cues, accelerating cascade formation.

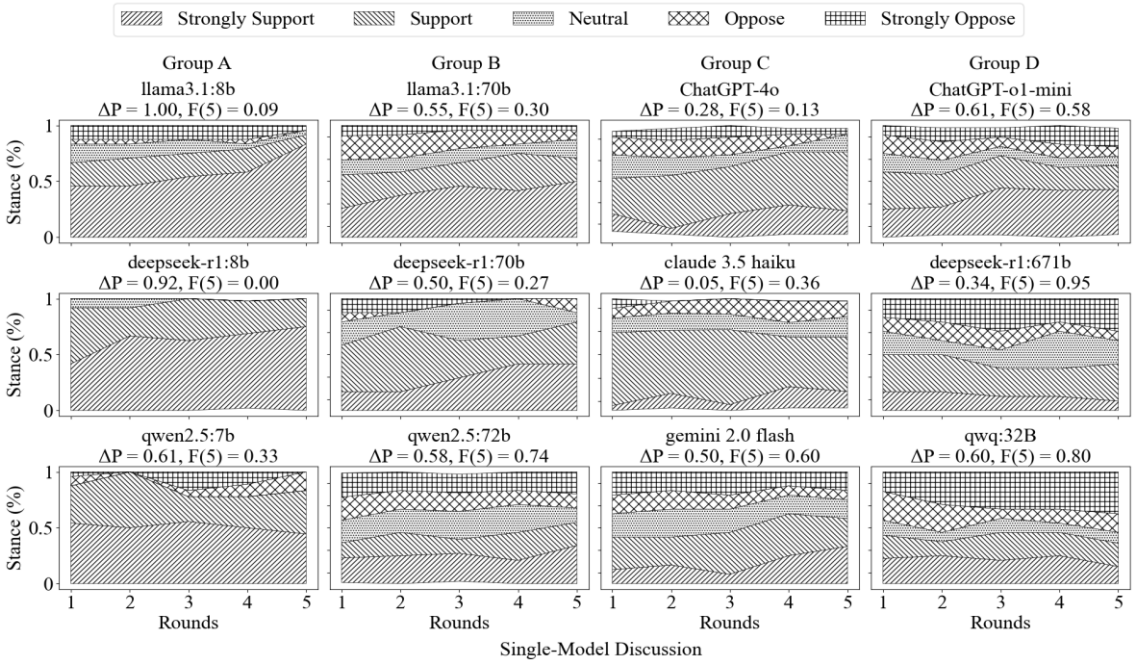


Fig. 3. Agent Stance Trends in Single-Group Discussions

The stance-evolution panels in Fig. 3 show the behavioral consequences of this alignment pressure. Several small and mid-sized Group A, B models display large increases in ΔP over the five rounds, often accompanied by a thinning neutral segment and a dominant pro-side by Round 5. In contrast, reasoning-oriented models register modest ΔP yet high final fragmentation F_5, preserving two sizable camps despite repeated exposure. Proprietary generalists occupy the middle of this spectrum: they polarize less aggressively than base models but do not preserve dissent as robustly as the strongest reasoning systems. This decoupling between ΔP and F_5, high ΔP with low F_5 for base models versus low-to-moderate ΔP with high F_5 for reasoning models: polarization captures movement toward extremes, while fragmentation captures structural two-camp persistence. Methodologically, these outcomes validate the five-round bulletin-board protocol as sensitive to both escalatory reinforcement ΔP and pluralism retention (F_5).

In synthesis, demonstrate that homogeneous rooms of a single model tend to self-reinforce: conformity rises, polarization often accelerates, and fragmentation collapses when base generative models dominate. By contrast, models with explicit reasoning modules act as in-room stabilizers—they resist majority pressure, limit outward drift (small ΔP), and preserve minority presence (high F_5).

4.2 Intra-Group Discussion Results

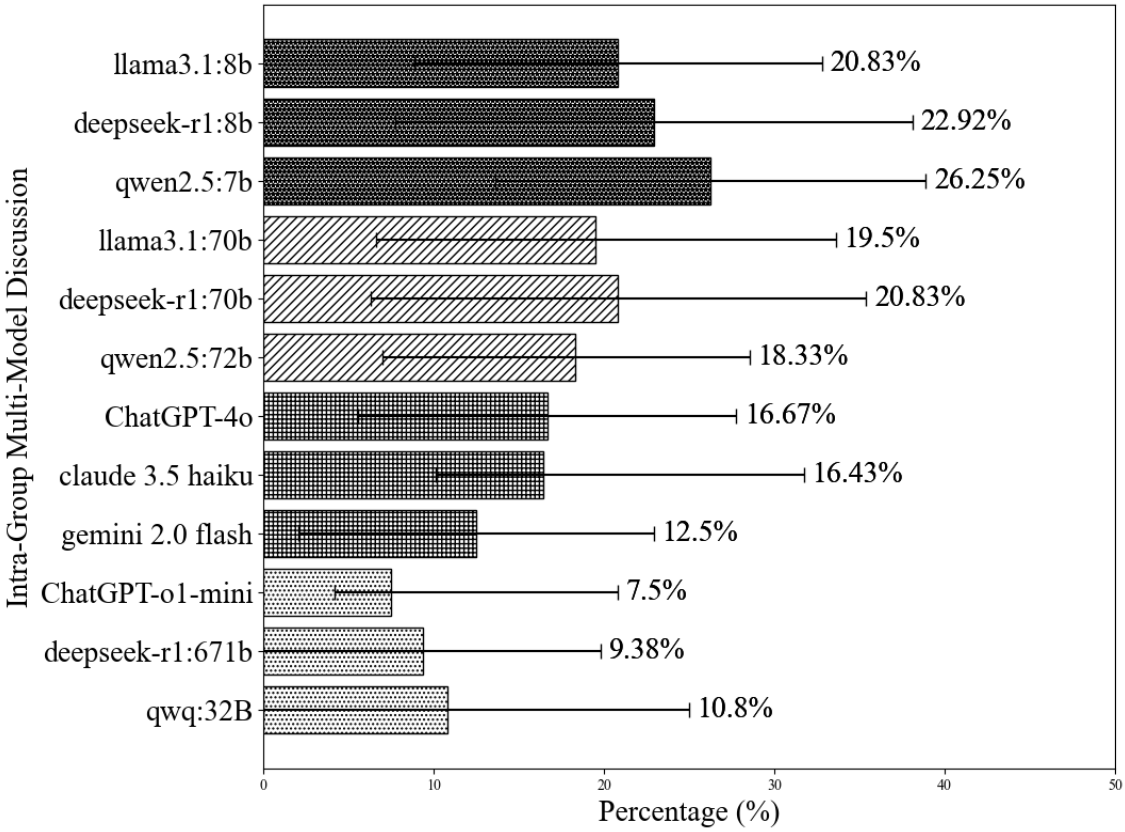


Fig. 4. Conformity Rates in Intra-Group Discussions.

As shown in Fig. 4, mixing models within the same family systematically alters alignment dynamics relative to the single-model baseline. Conformity rises notably for Groups A and B, indicating that within-family heterogeneity strengthens majority signals rather than regularizing them. Peaks around one quarter of available update opportunities illustrate that once a local majority forms, co-trained models tend to accommodate it. By contrast, Group D maintains single-digit to low-teens CR, consistent with reasoning-augmented architectures that preserve internal consistency and resist peer pressure.

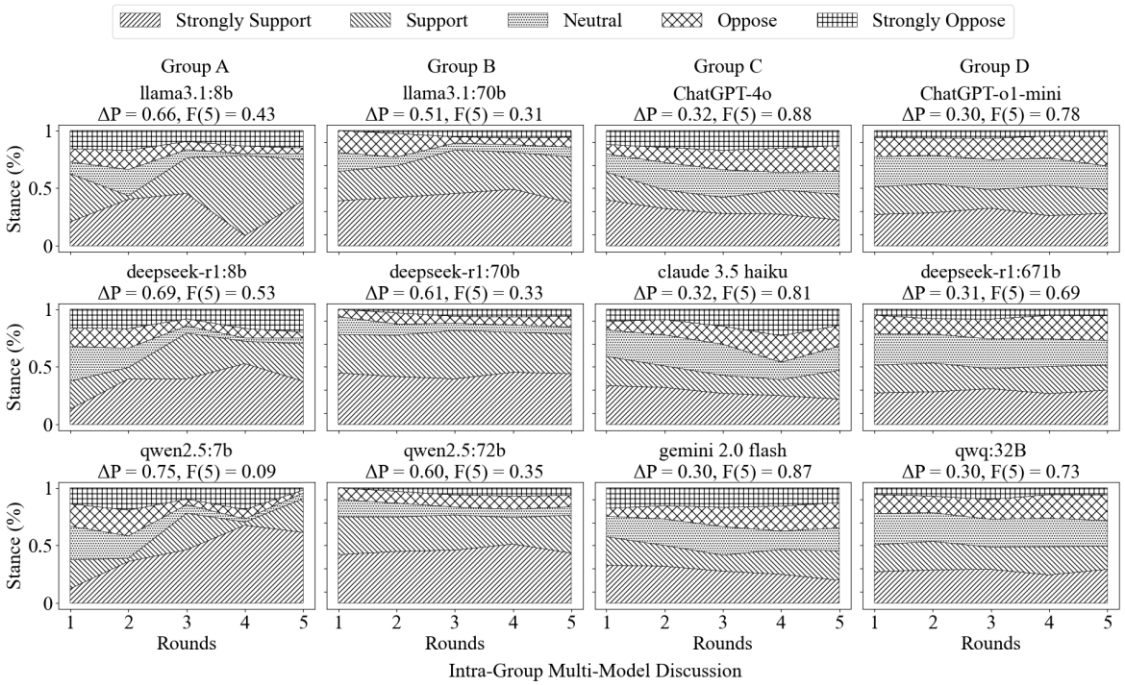


Fig. 5. Agent Stance Trends in Intra-Group Discussions.

The stance-evolution panels in Fig. 5 reveal the downstream consequences of this alignment pressure. Groups A and B display substantial increases in ΔP across rounds, with support segments expanding and neutral mass thinning—an interaction pattern consistent with repeated exposure among agents that share priors and rhetorical habits. In Groups C and D, ΔP is comparatively small, yet end states often retain two sizable camps, yielding high final fragmentation F_5 despite limited outward drift. This decoupling of polarization and fragmentation, low ΔP but high F_5, suggests that stronger deliberative scaffolding and more diverse inductive biases create stable dissent rather than escalation.

4.3 Cross-Group Discussion Results

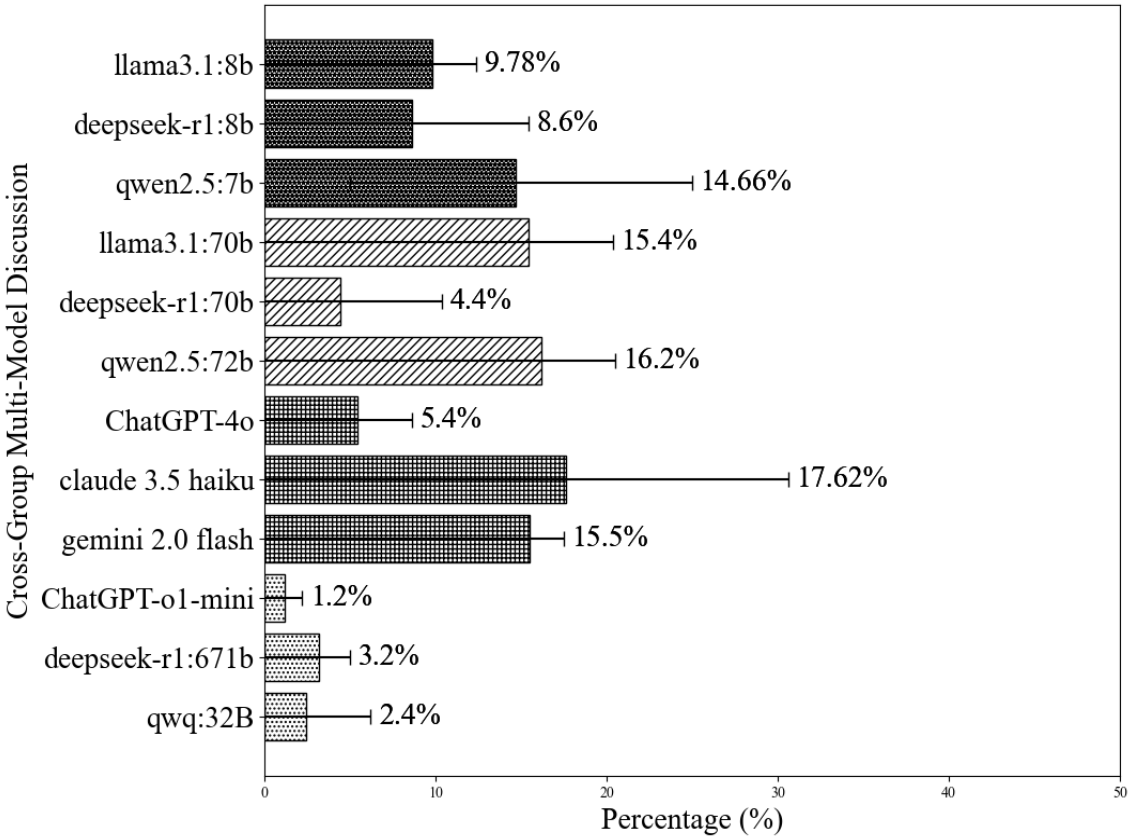


Fig. 6. Conformity Rates in Cross-Group Discussions.

As shown in Fig. 6, introducing cross-group heterogeneity, sampling agents from different model families with rotated roles across replications, reconfigures alignment dynamics relative to both the single-model and intra-group settings. Conformity is compressed toward lower levels for most participants, indicating that architectural diversity dilutes majority salience and inhibits cascade formation under our CR definition. Reasoning-oriented models remain the most resistant, while a subset of proprietary generalists displays mid-teens CR, suggesting that rhetorical fluency can still attract alignment even in heterogeneous rooms. Because composition is re-sampled and model-to-agent mappings are rotated, these differences are unlikely to be artifacts of seating or sampling; they reflect genuine architecture-by-context interactions under the replication protocol.

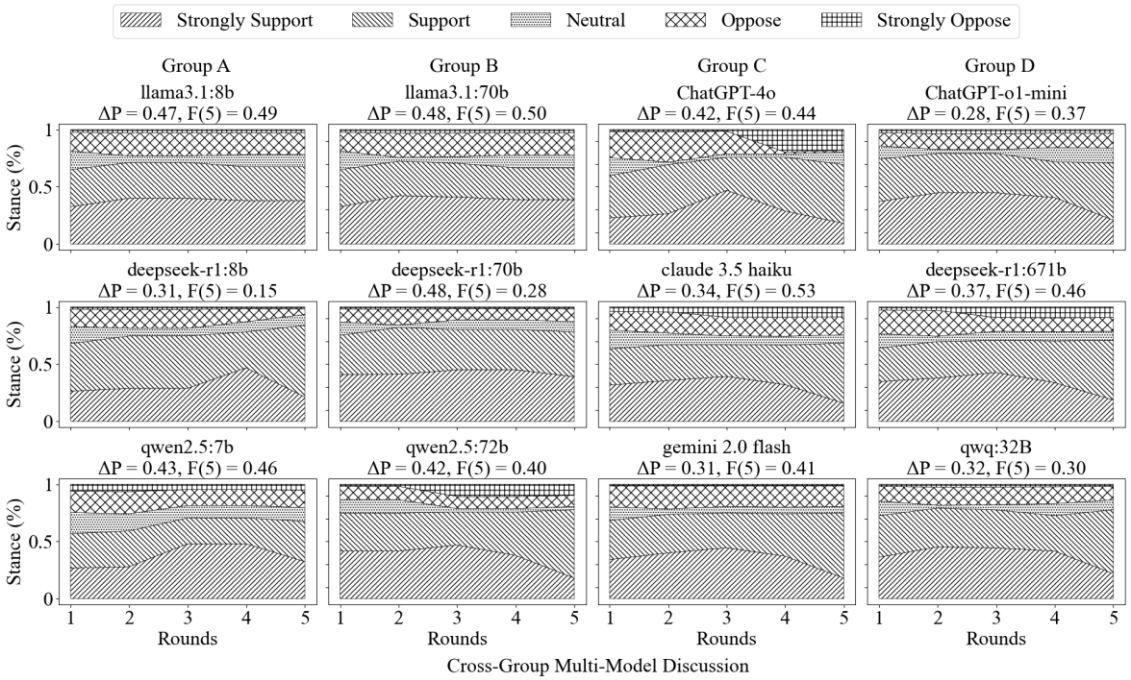


Fig. 7. Agent Stance Trends in Cross-Group Discussions.

The stance-evolution panels in Fig. 7 show the behavioral consequences of this conformity compression. Across families, polarization growth is modest ΔP tends to cluster in a narrow band, and the neutral mass is more persistent than in the intra-group setting. Nevertheless, final configurations often maintain two recognizable camps, indicating stable pluralism rather than convergence. Notably, specific mixtures momentarily exhibit rhetorical surges, brief expansions of a support segment in mid-rounds, yet these surges typically dissipate by Round 5, consistent with the idea that diversity of priors and argument styles reduces the durability of local majorities.

First, architectural diversity functions as a built-in regularizer on social influence: it suppresses conformity and decelerates polarization, even without changing the turn-taking protocol or personas. Second, the system does not simply converge to the center; instead, it stabilizes dissent, higher F_5 without large ΔP, suggesting that heterogeneous agents check each other's arguments rather than amplify them. Third, the contrast with intra-group results indicates that cross-family mixing is materially more effective than

within-family mixing at preventing majority cascades. For design, these findings recommend cross-model ensembles when the objective is to preserve viewpoint diversity and mitigate echo-chamber effects, and caution that homogeneity, even with multiple instances, can promote alignment pressure. Methodologically, the cross-group setting demonstrates that composition policy is a first-class control knob in LLM-based social simulations, on par with model scale and reasoning modules defined earlier.

4.4 Implications for Multi-Agent System Design

Across the three settings, a consistent pattern emerges: architecture and composition fundamentally govern social influence in LLM-based multi-agent systems. In single-model discussions, homogeneity amplifies majority pressure—conformity rises, polarization escalates, and minority stances collapse. Intra-group diversity does not inherently regularize these dynamics; for baseline generative families, it often intensifies cascades, whereas reasoning-oriented models preserve stability. Only in cross-group mixtures do conformity levels compress, polarization growth slow, and dissent persist through higher end-state fragmentation. These results confirm that composition policy is a first-order determinant of collective behavior, and that explicit reasoning capabilities—not scale alone—anchor deliberative robustness.

The implications are twofold. First, architecture must guide system selection: homogeneous generative models deliver rapid consensus, while reasoning-centric models are necessary for resisting majority cascades. Second, ensemble composition functions as a safety lever: “thin diversity” within families rarely prevents cascades, whereas “thick diversity” across families reliably moderates alignment pressure without reducing participation. Accordingly, a pre-deployment protocol should apply the three-tier harness—single-model, intra-group, and cross-group—under standardized replication. The triplet of metrics {CR, ΔP, F_5} yields a social dynamics signature that diagnoses

whether a composition risks echo-chamber amplification or sustains pluralism. Such diagnostics are model-agnostic, enabling behavioral stability to stand alongside accuracy and cost as a core deployment criterion.

Beyond architectural and compositional effects, a further implication concerns the interaction between model characteristics and persona design. Our findings indicate that the inherent properties of certain LLMs can outweigh externally assigned persona instructions. In particular, reasoning-oriented models in Group D consistently exhibited tendencies—such as extended analytical reasoning and argumentative depth—that persisted even when their personas prescribed alternative roles (e.g., skeptical or passive stances). This suggests that persona conditioning may not fully constrain model behavior when strong architectural biases are present.

For the design of multi-agent systems, this highlights an important methodological consideration: while persona design is a common strategy to induce heterogeneity, it cannot be assumed to operate independently of model-level characteristics. In practice, the interaction between model defaults and persona prompts may determine the trajectory of group dynamics more strongly than either factor alone. System designers should therefore account for these interactions, especially when seeking to balance diversity and control in large-scale simulations. Future work should explore systematic approaches to disentangling model-specific biases from persona effects, as this distinction has direct implications for both the validity and interpretability of LLM-based social simulations.

5. Conclusion

Building on these observations, we distill the design principles, experimental insights, and broader implications of our study. We introduced a structured simulation framework for examining social influence in LLM-based multi-agent systems, combining a five-round bulletin-board protocol, persona-grounded prompting, and a replication strategy that supports reliable inference. Our metric triplet, CR, ΔP, and the Fragmentation Index, provides complementary views of alignment pressure, movement toward extremes, and the persistence of opposing camps, yielding an interpretable "social dynamics signature" for any model or ensemble.

Empirically, the framework shows that baseline generative models tend to align quickly with perceived majorities, elevating CR and often accelerating ΔP, whereas reasoning-augmented architectures resist peer pressure and better preserve opinion diversity across rounds. These results underscore that architectural commitments, rather than scale alone, govern collective behavior in agentic settings and should therefore anchor model selection and deployment decisions.

Several limitations qualify these conclusions. The bulletin-board setting abstracts away turn overlap, asynchronous timing, and nonverbal cues; the simulations are text-only; and the behavioral metrics rely on discretized stance categories that may understate fine-grained shifts. Uniform persona and prompt templates may also interact with model-specific instruction adherence.

Future work should extend the protocol to longer horizons, incorporate multimodal and networked interactions, and complement our discrete metrics with continuous measures to more fully characterize the dynamics of influence. In addition, because the framework is inherently compatible with human participation, a promising

avenue lies in integrating human actors alongside LLM agents. Such hybrid settings would enable the study of human-computer interaction under controlled conditions, offering new insights into whether LLM-driven influence dynamics converge with, diverge from, or reshape patterns observed in human-only groups. Exploring these mixed-agent environments would provide stronger evidence for the external validity of LLM-based simulations and thereby positioning LLM-based social simulations as a methodological bridge between computational modeling and human behavioral research.

Reference

Aher, G., Arriaga, R. I., & Kalai, A. T. (2023). Using large language models to simulate multiple humans and replicate human subject studies. *Proceedings of the 40th International Conference on Machine Learning*, *202*, 337–371.

Asch, S. E. (1955). Opinions and social pressure. *Scientific American*, *193*(5), 31–35.

Cau, E., Pansanella, V., Pedreschi, D., & Rossetti, G. (2025). *Language-Driven Opinion Dynamics in Agent-Based Simulations with LLMs* (No. arXiv:2502.19098). arXiv. https://doi.org/10.48550/arXiv.2502.19098

Chuang, Y.-S., Goyal, A., Harlalka, N., Suresh, S., Hawkins, R., Yang, S., Shah, D., Hu, J., & Rogers, T. (2024). Simulating Opinion Dynamics with Networks of LLM-based Agents. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Findings of the Association for Computational Linguistics: NAACL 2024* (pp. 3326–3346). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.findings-naacl.211

DeepSeek-AI, Guo, D., Yang, D., Zhang, H., Song, J., Zhang, R., Xu, R., Zhu, Q., Ma, S., Wang, P., Bi, X., Zhang, X., Yu, X., Wu, Y., Wu, Z. F., Gou, Z., Shao, Z., Li, Z., Gao, Z., … Zhang, Z. (2025). *DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning* (No. arXiv:2501.12948). arXiv. https://doi.org/10.48550/arXiv.2501.12948

Flache, A., Mäs, M., Feliciani, T., Chattoe-Brown, E., Deffuant, G., Huet, S., & Lorenz, J. (2017). Models of Social Influence: Towards the Next Frontiers. *Journal of Artificial Societies and Social Simulation*, *20*(4), 2. https://doi.org/10.18564/jasss.3521

Gao, C., Lan, X., Lu, Z., Mao, J., Piao, J., Wang, H., Jin, D., & Li, Y. (2023). *S3:*

Social-network Simulation System with Large Language Model-Empowered Agents (No. arXiv:2307.14984). arXiv. https://doi.org/10.48550/arXiv.2307.14984

Grattafiori, A., Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Vaughan, A., Yang, A., Fan, A., Goyal, A., Hartshorn, A., Yang, A., Mitra, A., Sravankumar, A., Korenev, A., Hinsvark, A., … Ma, Z. (2024). *The Llama 3 Herd of Models* (No. arXiv:2407.21783). arXiv. https://doi.org/10.48550/arXiv.2407.21783

Guo, T., Chen, X., Wang, Y., Chang, R., Pei, S., Chawla, N. V., Wiest, O., & Zhang, X. (2024). *Large Language Model Based Multi-agents: A Survey of Progress and Challenges*. *9*, 8048–8057. https://doi.org/10.24963/ijcai.2024/890

Helbing, D. (Ed.). (2012). *Social Self-Organization: Agent-Based Simulations and Experiments to Study Emergent Social Behavior*. Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-24004-1

Introduction. (n.d.). CrewAI. Retrieved July 9, 2025, from https://docs.crewai.com/en/introduction

Jiang, Z., Shi, Y., Li, M., Xiao, H., Qin, Y., Wei, Q., Wang, Y., & Zhang, Y. (2024). *Casevo: A Cognitive Agents and Social Evolution Simulator* (No. arXiv:2412.19498). arXiv. https://doi.org/10.48550/arXiv.2412.19498

Jost, J. T., Baldassarri, D. S., & Druckman, J. N. (2022). Cognitive–motivational mechanisms of political polarization in social-communicative contexts. *Nature Reviews Psychology*, *1*(10), 560–576. https://doi.org/10.1038/s44159-022-00093-5

LangGraph. (n.d.). Retrieved July 9, 2025, from https://langchain-ai.github.io/langgraph/?utm_source=chatgpt.com

Leng, Y., & Yuan, Y. (2024). *Do LLM Agents Exhibit Social Behavior?* (No. arXiv:2312.15198). arXiv. https://doi.org/10.48550/arXiv.2312.15198

Li, C., Su, X., Han, H., Xue, C., Zheng, C., & Fan, C. (2023). *Quantifying the Impact of Large Language Models on Collective Opinion Dynamics* (No. arXiv:2308.03313). arXiv. https://doi.org/10.48550/arXiv.2308.03313

Liu, J., Huang, S., Aden, N., Johnson, N., & Song, C. (2023). Emergence of polarization in coevolving networks. *Physical Review Letters*, *130*(3), 037401. https://doi.org/10.1103/PhysRevLett.130.037401

Milgram, S. (1963). Behavioral Study of obedience. *The Journal of Abnormal and Social Psychology*, *67*(4), 371–378. https://doi.org/10.1037/h0040525

Minh Pham, T., Kondor, I., Hanel, R., & Thurner, S. (2020). The effect of social balance on social fragmentation. *Journal of The Royal Society Interface*, *17*(172), 20200752. https://doi.org/10.1098/rsif.2020.0752

Mou, X., Ding, X., He, Q., Wang, L., Liang, J., Zhang, X., Sun, L., Lin, J., Zhou, J., Huang, X., & Wei, Z. (2024). *From Individual to Society: A Survey on Social Simulation Driven by Large Language Model-based Agents* (No. arXiv:2412.03563). arXiv. https://doi.org/10.48550/arXiv.2412.03563

OpenAI, Hurst, A., Lerer, A., Goucher, A. P., Perelman, A., Ramesh, A., Clark, A., Ostrow, A. J., Welihinda, A., Hayes, A., Radford, A., Mądry, A., Baker-Whitcomb, A., Beutel, A., Borzunov, A., Carney, A., Chow, A., Kirillov, A., Nichol, A., … Malkov, Y. (2024). *GPT-4o System Card* (No. arXiv:2410.21276). arXiv. https://doi.org/10.48550/arXiv.2410.21276

Piao, J., Lu, Z., Gao, C., Xu, F., Santos, F. P., Li, Y., & Evans, J. (2025). *Emergence of human-like polarization among large language model agents* (No. arXiv:2501.05171). arXiv. https://doi.org/10.48550/arXiv.2501.05171

Piao, J., Yan, Y., Zhang, J., Li, N., Yan, J., Lan, X., Lu, Z., Zheng, Z., Wang, J. Y., Zhou, D., Gao, C., Xu, F., Zhang, F., Rong, K., Su, J., & Li, Y. (2025). *AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents Advances Understanding of Human Behaviors and Society* (No. arXiv:2502.08691; Version 1). arXiv. https://doi.org/10.48550/arXiv.2502.08691

Qasmi, A., Naseem, U., & Nasim, M. (2025). *Competing LLM Agents in a Non-Cooperative Game of Opinion Polarisation* (No. arXiv:2502.11649). arXiv. https://doi.org/10.48550/arXiv.2502.11649

Qwen, Yang, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Li, C., Liu, D., Huang, F., Wei, H., Lin, H., Yang, J., Tu, J., Zhang, J., Yang, J., Yang, J., Zhou, J., Lin, J., … Qiu, Z. (2025). *Qwen2.5 Technical Report* (No. arXiv:2412.15115). arXiv. https://doi.org/10.48550/arXiv.2412.15115

Squazzoni, F., Jager, W., & Edmonds, B. (2014). Social Simulation in the Social Sciences: A Brief Overview. *Social Science Computer Review*, *32*(3), 279–294. https://doi.org/10.1177/0894439313512975

Team, G., Anil, R., Borgeaud, S., Alayrac, J.-B., Yu, J., Soricut, R., Schalkwyk, J., Dai, A. M., Hauth, A., Millican, K., Silver, D., Johnson, M., Antonoglou, I., Schrittwieser, J., Glaese, A., Chen, J., Pitler, E., Lillicrap, T., Lazaridou, A., … Vinyals, O. (2024). *Gemini: A Family of Highly Capable Multimodal Models* (No. arXiv:2312.11805). arXiv. https://doi.org/10.48550/arXiv.2312.11805

Team, Q. (2024, November 28). *QwQ: Reflect Deeply on the Boundaries of the Unknown*. Qwen. https://qwenlm.github.io/blog/qwq-32b-preview/

The group effect: Social cohesion and health outcomes. (n.d.). ResearchGate. Retrieved September 2, 2025, from

https://www.researchgate.net/publication/289371828_The_group_effect_Social_cohesion_and_health_outcomes

Wang, L., Zhang, Z., & Chen, X. (2025). *Investigating and Extending Homans' Social Exchange Theory with Large Language Model based Agents* (No. arXiv:2502.12450). arXiv. https://doi.org/10.48550/arXiv.2502.12450

Wang, Z., Mao, S., Wu, W., Ge, T., Wei, F., & Ji, H. (2024). Unleashing the Emergent Cognitive Synergy in Large Language Models: A Task-Solving Agent through Multi-Persona Self-Collaboration. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (pp. 257–279). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.naacl-long.15

Weng, Z., Chen, G., & Wang, W. (2025). *Do as We Do, Not as You Think: The Conformity of Large Language Models* (No. arXiv:2501.13381). arXiv. https://doi.org/10.48550/arXiv.2501.13381

Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., Awadallah, A. H., White, R. W., Burger, D., & Wang, C. (2023). *AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation* (No. arXiv:2308.08155). arXiv. https://doi.org/10.48550/arXiv.2308.08155

yet, A. A. P. N. published yet D. N. D. (n.d.). *On the Biology of a Large Language Model*. Transformer Circuits. Retrieved April 13, 2025, from https://transformer-circuits.pub/2025/attribution-graphs/biology.html

Zhang, J., Xu, X., Zhang, N., Liu, R., Hooi, B., & Deng, S. (2024). Exploring Collaboration Mechanisms for LLM Agents: A Social Psychology View. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd Annual*

Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (pp. 14544–14607). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.782

Zhong, T., Liu, Z., Pan, Y., Zhang, Y., Zhou, Y., Liang, S., Wu, Z., Lyu, Y., Shu, P., Yu, X., Cao, C., Jiang, H., Chen, H., Li, Y., Chen, J., Hu, H., Liu, Y., Zhao, H., Xu, S., … Liu, T. (2024). *Evaluation of OpenAI o1: Opportunities and Challenges of AGI* (No. arXiv:2409.18486). arXiv. https://doi.org/10.48550/arXiv.2409.18486